

# FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

## Student Assessment Document

### FINAL EXPLANATION: GENERAL SCIENCE – GRADE 10 | SUB-STRAND 1.2: THE CELL

#### Student Assessment Document

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| Student Name | _____ |
| Class        | _____ |
| Date         | _____ |

### INSTRUCTIONS FOR STUDENTS

This is your Final Explanation task. Throughout this unit, you have investigated the driving question: **What makes up an organism — and how do the tiny structures inside a single cell keep a whole living thing, from a sukuma wiki plant to a human being in Nairobi, alive?**

You started by looking at a sukuma wiki leaf under a light microscope in Kisumu, and you ended up exploring the ultra-fine internal world of cells revealed by electron microscopy. Now it is time to bring everything you have learned together into one well-organised, evidence-based explanation.

**Your task:** Write a full, detailed explanation that answers the driving question. Your explanation must:

- Move logically from organism → organ system → organ → tissue → cell → organelle
- Explain why the light microscope and the electron microscope show different levels of detail
- Identify and explain the function of at least **six organelles** found in a plant cell (using the sukuma wiki leaf as your example) and at least **four organelles** found in a human cell
- Connect each organelle's function to the survival of the whole organism
- Use scientific vocabulary correctly throughout
- Include at least **one labelled diagram** of a plant cell and one of an animal cell

Use the section prompts below to structure your response. Write in full sentences and paragraphs — bullet points alone are not enough. Imagine you are explaining to a fellow Grade 10 learner in Nairobi who missed the entire unit and needs to understand everything from scratch.

### Section 1: Levels of Organisation — From Sukuma Wiki to You

**Explain the levels of biological organisation, starting from the cell and ending at the organism. Use the sukuma wiki plant AND the human body as your two examples. For each level, name**

All living things — from the sukuma wiki plants grown along the shores of Lake Victoria to the students eating githeri in a Nairobi school canteen — are built from the same fundamental unit: the cell. Cells are the smallest units of life capable of carrying out all basic life processes on their own.

Cells of the same type group together to form tissues. In a sukuma wiki leaf, for example, mesophyll cells packed with green

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| <p><b>the level, give one specific example from each organism, and explain how that level depends on the level below it. Why can't a sukuma wiki plant or a human being exist without cells?</b></p> | <p>chloroplasts form photosynthetic tissue whose job is to trap sunlight and make food. In the human body, muscle cells join together to form muscle tissue, which can contract and produce movement.</p> <p>Tissues of different types are then organised into organs — structures that perform a specific function. The sukuma wiki leaf is an organ: it contains photosynthetic tissue, vascular tissue (which carries water and sugars), and epidermal tissue (which protects the leaf surface). In a human, the stomach is an organ made of muscular tissue, glandular tissue, and epithelial tissue working together to digest food.</p> <p>Organs that work together on a shared function form an organ system. In the human body, the stomach, small intestine, liver, and large intestine all belong to the digestive system. In a flowering plant such as sukuma wiki, the roots, stem, and leaves form the shoot and root systems that together absorb, transport, and process nutrients.</p> <p>Finally, all the organ systems together make up an organism — the complete, living individual. Neither the sukuma wiki plant nor a human being in Nairobi could survive if the cellular level broke down, because every life process — from releasing energy to responding to the environment — is ultimately carried out inside individual cells. When the sukuma wiki leaf is cut from the plant and left in the sun, its cells can no longer receive water through the vascular system; they lose water by evaporation, the internal machinery shuts down, and the leaf dies. This shows that life depends on the continuous, coordinated activity of cells organised into higher levels.</p> |
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## Section 2: Two Microscopes, Two Worlds — Resolving the Puzzle

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| <p><b>Explain why the light microscope and the electron microscope gave the Kisumu learner such different images of the same sukuma wiki cell. In your answer, define magnification and resolution, explain how each type of microscope works (in simple terms), state the approximate maximum resolution of each, and explain what this means for the level of detail each microscope can reveal. Which organelles can you see with a light microscope, and which require an electron microscope? Why?</b></p> | <p>When the learner in Kisumu looked at the sukuma wiki leaf under the school's light microscope, they could clearly see the box-like cell walls and the green chloroplasts inside each cell — but the image stopped there. When the teacher showed a photomicrograph taken with an electron microscope, far more was visible: folded internal membranes inside the chloroplasts, the double membrane of the nucleus, the intricate folds of the mitochondria (cristae), and the network of the endoplasmic reticulum. Why did the two microscopes reveal such different pictures of the same cell?</p> <p>The answer lies in two key ideas: magnification and resolution. Magnification tells you how many times larger an image appears compared to the real object. Resolution (also called resolving power) tells you how much detail you can actually distinguish — specifically, the smallest distance between two points that still appear as two separate points rather than blurring into one.</p> <p>A school light microscope uses visible light to illuminate the specimen. Because the wavelength of visible light is relatively large (roughly 400–700 nanometres), the best possible resolution a light microscope can achieve is about 200 nm (0.2 micrometres). This is good enough to see cells, cell walls, chloroplasts, and the nucleus as a dark blob — but the internal membranes of organelles are far smaller than 200 nm and simply blur together into an invisible smudge.</p> <p>An electron microscope uses a beam of electrons instead of light. Electrons have a wavelength thousands of times smaller than visible light, giving the electron microscope a resolution of about 0.1–0.2 nm — roughly 1,000 times better than a light microscope. This is why the electron microscope reveals the folded inner membranes of chloroplasts (thylakoids), the cristae of mitochondria, ribosomes dotting the rough endoplasmic reticulum, and the fine structure of the nuclear envelope.</p> <p>In summary: both microscopes were looking at the same sukuma wiki cell, but the light microscope could only show structures</p> |
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larger than about 200 nm (cell wall, vacuole, chloroplasts, nucleus), while the electron microscope could reveal structures as small as a fraction of a nanometre. The apparent difference in the two images was not a contradiction — it was a difference in resolving power.

### Section 3: Inside the Sukuma Wiki Cell — Plant Cell Organelles and Their Functions

**Describe the structure and function of at least six organelles or structures found in a plant cell, using the sukuma wiki leaf cell as your model. For each organelle, explain: (a) what it looks like, (b) what it does, and (c) how its function contributes to keeping the sukuma wiki plant alive. Include your labelled diagram of a plant cell here. Remember to include organelles unique to plant cells as well as those shared with animal cells.**

A fresh sukuma wiki leaf cell, when viewed with an electron microscope, is a busy, highly organised space. Every structure inside the cell has a specific job that keeps the plant alive. Here are the key organelles and structures you would find:

**\*\*Cell wall:\*\*** The outermost layer of a plant cell is the rigid cell wall, made mainly of cellulose fibres. It gives the cell a fixed, box-like shape — which is exactly what the Kisumu learner saw as 'tiny box-like compartments' under the light microscope. The cell wall supports the leaf so it stands upright and does not collapse, and it protects the cell from bursting when it absorbs water.

**\*\*Cell membrane (plasma membrane):\*\*** Just inside the cell wall is the flexible cell membrane, a thin double layer of lipid molecules with proteins embedded in it. It controls what enters and leaves the cell — letting in water, carbon dioxide, and minerals while keeping harmful substances out. Without this selective barrier, the cell's internal chemistry would fall apart.

**\*\*Chloroplasts:\*\*** The green granules the learner saw are chloroplasts — organelles found only in plant cells. Each chloroplast has a double outer membrane and, inside, a system of flattened, disc-like membranes called thylakoids, which are stacked into grana. The chloroplasts contain the green pigment chlorophyll, which captures sunlight energy. During photosynthesis, chloroplasts use this energy to convert carbon dioxide (from the air) and water (absorbed through the roots) into glucose — the food that keeps the sukuma wiki plant alive and also nourishes the person who eats it.

**\*\*Nucleus:\*\*** The large, oval, membrane-bound nucleus is the control centre of the cell. It contains the cell's DNA, organised into chromosomes, which carry the genetic instructions for every protein the cell needs to make. The nucleus directs all cell activities, including when to grow, when to divide, and which enzymes to produce for photosynthesis and respiration.

**\*\*Mitochondria:\*\*** Although chloroplasts make glucose, the cell needs energy in a usable form — ATP. Mitochondria are oval organelles with a folded inner membrane (the cristae) that provides a large surface area for the chemical reactions of aerobic respiration. They break down glucose from photosynthesis and release energy as ATP, which powers every other process in the cell, including active transport across the cell membrane.

**\*\*Vacuole:\*\*** Plant cells have a large central vacuole filled with cell sap (a watery solution of sugars, salts, and pigments). In a healthy, well-watered sukuma wiki leaf, the vacuole is fully inflated and pushes the cytoplasm outward against the cell wall, creating turgor pressure that keeps the leaf firm and upright. When the leaf is left in the sun without water, the vacuole shrinks, turgor pressure is lost, and the leaf wilts — exactly what the learner observed.

**\*\*Endoplasmic reticulum (ER) and Ribosomes:\*\*** The rough endoplasmic reticulum is a network of folded membranes studded with tiny ribosomes. Ribosomes are the sites where proteins are assembled according to the instructions from the nucleus. These proteins include chlorophyll-binding proteins, enzymes for photosynthesis, and structural proteins for the cell wall. The smooth ER (without ribosomes) is involved in lipid production and transport.

\*[Diagram: A labelled drawing of a sukuma wiki plant cell should appear here, showing the cell wall, cell membrane, chloroplasts with thylakoids, nucleus with nuclear envelope and nucleolus, large central vacuole, mitochondria, rough ER, ribosomes, and Golgi apparatus.]\*

## Section 4: The Human Cell — Similarities, Differences, and Survival

**Compare the human (animal) cell to the sukuma wiki (plant) cell. Describe at least four organelles in a human cell, explaining each one's function and how it keeps a person in Nairobi alive. Clearly state THREE structures present in the plant cell but absent in a typical human cell, and explain why this difference makes sense given how each organism obtains energy and maintains its shape. Include your labelled diagram of an animal cell here.**

Human cells — for example, the cells lining your intestine as they absorb nutrients from a plate of githeri — share many organelles with sukuma wiki cells, but there are also important differences that reflect the very different lifestyles of animals and plants.

**\*\*Shared organelles in human cells:\*\***

**\*\*Nucleus:\*\*** Just as in the sukuma wiki cell, the human cell's nucleus contains DNA and directs all cell activities. In intestinal cells, the nucleus switches on genes for digestive enzymes; in muscle cells of a Kenyan athlete running at the Nyayo Stadium, it activates genes for contractile proteins.

**\*\*Mitochondria:\*\*** Human cells cannot photosynthesise — they must obtain glucose by eating food. The mitochondria in human cells carry out aerobic respiration, breaking down glucose (from ugali, beans, or sukuma wiki) to produce ATP. Active cells, such as heart muscle cells, contain hundreds of mitochondria because they need large amounts of energy continuously.

**\*\*Ribosomes and Rough Endoplasmic Reticulum:\*\*** Human cells also make proteins using ribosomes on the rough ER. These proteins include enzymes, hormones (such as insulin), and antibodies that protect against disease — all essential for survival.

**\*\*Cell membrane:\*\*** The human cell membrane, like the plant cell membrane, is a phospholipid bilayer that controls what enters and leaves the cell. It also allows cells to communicate with each other through receptor proteins, which is critical for coordinating the body's responses — for example, when blood sugar rises after eating nyama choma, receptor proteins on liver cells detect insulin and trigger the uptake of glucose.

**\*\*Structures present in plant cells but ABSENT in typical human cells:\*\***

1. **\*\*Cell wall\*\*** — Human cells have no cell wall. Animal cells get their shape from the internal cytoskeleton and are supported by surrounding tissues and bones. Without a rigid wall, human cells can change shape — essential for processes like white blood cells engulfing bacteria, or red blood cells squeezing through narrow capillaries.
2. **\*\*Chloroplasts\*\*** — Human cells cannot photosynthesise; we depend entirely on eating plants (or animals that ate plants) for our energy. There is no need for chloroplasts in human cells.
3. **\*\*Large central vacuole\*\*** — Human cells may have small vacuoles, but they lack the large, permanent central vacuole of plant cells. Plant cells need this structure for turgor pressure and storage; human cells maintain their fluid balance differently, through the kidneys and the cell membrane.

\*[Diagram: A labelled drawing of a human (animal) cell should appear here, showing the cell membrane, nucleus with nuclear

envelope, mitochondria, rough and smooth ER, ribosomes, Golgi apparatus, lysosomes, and cytoplasm. Note the absence of cell wall, chloroplasts, and large central vacuole.]\*

## Section 5: Answering the Driving Question — Connecting Cells to the Living Organism

**Now bring it all together. Write a concluding explanation that directly answers the driving question: 'What makes up an organism — and how do the tiny structures inside a single cell keep a whole living thing, from a sukuma wiki plant to a human being in Nairobi, alive?' Your conclusion must: (a) summarise the levels of organisation, (b) explain how the organelles you described work together as a system — not in isolation — to sustain life, (c) connect the phenomenon (the sukuma wiki leaf wilting and dying) back to the failure of cellular structures, and (d) reflect on why understanding cells matters for real life in Kenya — think about food, health, farming, or medicine.**

You began this unit puzzled by a simple observation: a sukuma wiki leaf left in the sun wilts, dries, and dies. After eight lessons of investigation — from peering through the school microscope in Kisumu to studying electron photomicrographs — you can now answer the driving question in full.

An organism is not a single, simple thing. It is a carefully organised hierarchy: cells form tissues, tissues form organs, organs form organ systems, and organ systems form the complete organism. Every level depends on the one below it, and the foundation of it all is the cell. Both the sukuma wiki plant and the human body in Nairobi are, at their most fundamental level, communities of cells — each cell a miniature factory running continuously to keep the whole organism alive.

Inside each cell, the organelles do not work in isolation; they work as an integrated system. In a sukuma wiki leaf cell, the chloroplasts capture sunlight and produce glucose. The mitochondria convert that glucose into ATP — the chemical currency of energy. The nucleus uses that energy to read DNA and direct the ribosomes to make the proteins needed for every structure and chemical reaction in the cell. The cell membrane carefully regulates what comes in and what goes out, while the vacuole maintains the turgor pressure that keeps the leaf firm. The cell wall gives each cell its shape and the strength to support the plant. Remove any one of these components and the whole system begins to fail.

This is exactly what happened to the wilting sukuma wiki leaf. When the leaf was separated from the plant, water could no longer travel up through the vascular system to reach the cells. Without water, the large central vacuoles in each leaf cell deflated. Turgor pressure collapsed, and the leaf drooped. With no water, photosynthesis in the chloroplasts stopped — no more glucose was produced. Without glucose, the mitochondria could not make ATP. Without ATP, the cell membrane could no longer carry out active transport, the nucleus could not maintain cell activities, and the cell's internal machinery ground to a halt. The leaf did not die all at once — it died organelle by organelle, process by process, cell by cell.

Understanding cells is not just a school subject — it has real importance for life in Kenya. Farmers in the Rift Valley who understand how plant cells use water can irrigate their crops more efficiently during dry seasons. Health workers in Mombasa who understand how human cells are attacked by pathogens can better explain disease and prevention to communities. Scientists and doctors who understand how cancer begins — when the nucleus loses control of cell division — can develop better treatments. Even understanding why sukuma wiki is nutritious comes down to cellular biology: the glucose, vitamins, minerals, and fibre stored in its cells nourish the cells of everyone who eats it.

In summary: an organism is made up of billions of cells, each one a fully functional unit of life. The structures inside each cell — the organelles — are not separate, unrelated parts. They are an interconnected system, and it is the continuous, coordinated activity of this system, multiplied across billions of cells and organised into tissues, organs, and organ systems, that keeps a living thing alive. From the sukuma wiki on a shamba in Kisumu to the student reading this page in Nairobi — life, in all its complexity, begins in the cell.

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| Criterion                                                               | Excellent (4)                                                                                                                                                                                                                                                                                                                               | Proficient (3)                                                                                                                                                                                                                                                                          | Developing (1–2)                                                                                                                                                                                                                           |
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| Levels of Biological Organisation                                       | Accurately and clearly explains all levels from cell to organism using both sukuma wiki and the human body as examples. Shows precise understanding of how each level depends on the one below it. Uses correct scientific vocabulary throughout (e.g., tissue, organ, organ system, organism).                                             | Explains most levels of organisation with mostly correct examples from both organisms. Minor gaps or one vocabulary error present, but the overall logic is clear and correct.                                                                                                          | Attempts to describe levels of organisation but confuses two or more levels, uses only one organism as an example, or has significant vocabulary errors. The relationship between levels is unclear.                                       |
| Microscopy — Magnification and Resolution                               | Clearly and accurately explains the difference between magnification and resolution, states approximate resolution limits for both light and electron microscopes, correctly identifies which organelles are visible with each type, and fully explains why the two microscopes produced such different images of the sukuma wiki cell.     | Explains magnification and resolution with mostly correct information. Correctly identifies the resolution difference between the two microscopes and gives a reasonable explanation of why the images differ. May omit one specific organelle example or approximate resolution value. | Attempts to explain the difference between the two microscopes but confuses magnification with resolution, or cannot accurately explain why the electron microscope shows more detail. Limited or incorrect use of scientific terminology. |
| Plant Cell Organelles — Structure, Function, and Connection to Survival | Accurately describes at least six plant cell organelles (including chloroplasts, cell wall, vacuole, nucleus, mitochondria, and cell membrane), explains the structure and function of each, and clearly connects each organelle's function to the survival of the sukuma wiki plant. Includes a clearly labelled diagram of a plant cell.  | Accurately describes at least five organelles with correct functions. The connection to plant survival is made for most organelles. Diagram is present and mostly correct, with minor labelling omissions.                                                                              | Describes fewer than four organelles, or has significant errors in the function of two or more organelles. Connection to plant survival is weak or absent. Diagram is missing, unlabelled, or significantly incorrect.                     |
| Comparison of Plant and Animal (Human) Cells                            | Accurately describes at least four human cell organelles and their functions, correctly identifies three structures present in plant cells but absent in human cells (cell wall, chloroplasts, large central vacuole), and gives a clear, logical biological reason for each difference. Includes a correctly labelled animal cell diagram. | Describes at least three human cell organelles correctly and identifies at least two structures unique to plant cells. Reasons for differences are mostly logical. Diagram is present with minor errors or omissions.                                                                   | Describes fewer than three human cell organelles, or identifies fewer than two plant-specific structures. Reasons for differences between cell types are incorrect or missing. Diagram is absent or significantly inaccurate.              |
| Driving Question — Synthesis and Real-Life Connection                   | The concluding section directly and fully answers the driving question, synthesises all previous sections coherently, clearly explains the wilting phenomenon in terms of organelle                                                                                                                                                         | The conclusion answers the driving question and connects most sections together. The wilting phenomenon is explained with reference to at least two organelles or processes. A real-life                                                                                                | The conclusion attempts to address the driving question but is incomplete or inconsistent with evidence presented in earlier sections. The explanation of the wilting phenomenon is vague or                                               |

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|  | failure, and makes a specific, insightful connection to real life in Kenya (farming, health, medicine, or nutrition). Writing is clear, logical, and uses scientific vocabulary accurately. | Kenyan connection is made, though it may be general rather than specific. Writing is mostly clear with minor errors. | incorrect. Little or no real-life connection is made. Writing is difficult to follow in places. |
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